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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CC3211-1- ADHOC WIRELESS NETWORKS

(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which routing protocol is reactive in ad hoc networks?
A) DSDV
B) OLSR
C) AODV
D) WRP
2. Which of the following protocols maintains routing information periodically?
A) DSR
B) AODV
C) DSDV
D) ZRP
3. The hidden terminal problem in ad hoc networks occurs when
A) Two nodes are within range but can't detect each other's transmission
B) Nodes are out of range of each other but interfere at a common receiver
C) The network is overloaded with hidden channels
D) Encryption hides the channel
4. A typical application of ad hoc wireless networks is
A) Cellular networks
B) Military battlefield communication
C) Fixed broadband
D) Optical fibre backbone
5. Which of the following attacks is common in ad hoc networks?
A) Blackhole attack
B) SQL injection
C) Password brute force
D) DNS poisoning
6. The main trade-off in ad hoc routing is between
A) Security and energy
B) Throughput and jitter
C) Routing overhead and delay
D) Scalability and encryption
7. What is the role of RTid in MARCH?
A) Avoids misinterpretation of CTS packets
B) Provides encryption
C) Acts as session ID
D) Used for error detection
8. Which issue does DSDV solve using sequence numbers?
A) Routing loops
B) Mobility prediction
C) Bandwidth allocation
D) Power consumption
9. WRP avoids the 'count-to-infinity' problem by
A) Using sequence numbers
B) Maintaining predecessor information
C) Flooding route requests
D) Clustering
10. LAR reduces routing overhead by
A) Flooding to all nodes
B) Restricting search to a request zone
C) Using cluster-heads
D) Broadcasting periodically
11. Which is a drawback of DSR?
A) Route cache may contain stale paths
B) Large routing tables
C) Energy inefficiency in cluster-heads
D) No support for mobility
12. The main idea of CEDAR is
A) Using a core set of nodes to perform routing
B) Using source routing for all packets
C) Relying on GPS-based positions
D) Flooding entire network for routes

13. ZRP uses proactive routing within
 - A) Zone
 - B) Cluster
 - C) Request zone
 - D) Region
14. Which metric minimizes energy consumed per packet?
 - A) Minimum energy routing
 - B) Maximum residual battery capacity
 - C) Load balancing
 - D) Minimum hop count
15. SEAD prevents which type of attack?
 - A) Routing table poisoning
 - B) Wormhole
 - C) Grayhole
 - D) Replay
16. Which of the following is not a primary network security requirement?
 - A) Confidentiality
 - B) Integrity
 - C) Scalability
 - D) Availability
17. Key management deals with
 - A) Encrypting and decrypting messages
 - B) Generating, distributing, and revoking cryptographic keys
 - C) Detecting routing loops
 - D) Avoiding congestion
18. Security provisioning in adhoc networks must balance
 - A) Performance, overhead, and security strength
 - B) Energy efficiency only
 - C) Bandwidth and topology changes only
 - D) Routing loops and latency only
19. A wormhole attack involves
 - A) Creating a tunnel between two attackers to disrupt routing
 - B) Flooding the network with HELLO packets
 - C) Modifying routing tables
 - D) Exhausting battery resources
20. In a Sybil attack, a node
 - A) Creates multiple fake identities
 - B) Modifies sequence numbers
 - C) Drops packets selectively
 - D) Forwards packets with delay

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is Destination Sequenced Distance-Vector Routing Protocol? Explain its working mechanism with examples.	8	L*2	1	2
	b) Mention and explain the Issues In Designing A Mac Protocol For Adhoc Wireless Networks.	6	L2	2	1
2.	a) What is Media Access with Reduced Handshake (MARCH) protocol. Explain its working in detail.	8	L2	1	2
	b) Explain the working model of MACAW protocol	8	L2	2	1,2
3.	a) Explain the Hop Reservation Multiple Access Protocol along with examples.	8	L2	1	1
	b) Explain the Busy Tone Multiple Access Protocol with examples.	8	L2	1	2
Unit – II					
4.	a) What is the Wireless Routing Protocol? Explain its working mechanism with examples.	8	L2	2	1,2
	b) What is Cluster-Head Gateway Switch Routing Protocol? Explain its working mechanism with examples.	8	L2	2	1,2
5.	a) What is the Dynamic Source Routing Protocol? Explain its working mechanism with examples.	8	L2	3	1,8
	b) What is the Ad Hoc On-Demand Distance-Vector Routing Protocol? Explain its working mechanism with examples.	8	L2	2	3

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| 6. a) | What is Adhoc TCP? Explain the working with an example. | 8 | L2 | 3 | 3,1 |
| b) | What are the issues In Designing A Transport Layer Protocol For Ad Hoc Wireless Networks? Explain each of them | 8 | L2 | 3 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | What are the issues and challenges in security provisioning for Adhoc Wireless Networks? | 8 | L2 | 5 | 3,1 |
| b) | Explain the Security-Aware Ad Hoc Routing Protocol with a scenario. | 8 | L2 | 4 | 3,2 |
| 8. a) | What is the Secure Efficient Ad Hoc Distance Vector Routing Protocol? Explain the working in detail. | 8 | L2 | 4 | 3,2 |
| b) | Explain the working of the Ticket-Based QoS Routing Protocol with an example. | 8 | L2 | 5 | 3,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
